Waveform Quality, Confidence, and AF-Case Retention in Single Lead ECG Selective Classification: A Controlled Noise Benchmark

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Abstract

Background: With selective prediction, an ECG classifier may withhold predictions expected to be erroneous. Despite achieving good aggregate coverage, poor coverage on the specific disease under detection could remain hidden. Objectives: We examined whether waveform quality metrics improve upon error ranking based on classifier probabilities and classes predicted when a noise contamination source withheld from development of methodology is introduced, retaining instances of atrial fibrillation (AF). Methods: Retrospective computational benchmark analyses of 15,292 PTB-XL human-reviewed, adult ECGs from 13,796 participants were conducted. Lead II signals were extracted from 10-second segments sampled at 100 Hz. Distinct roles based on participant separation were utilized in training a fixed classifier model and selecting checkpoints, fitting error-ranking models, determining a selective prediction strategy on unlabelled data, and finally testing. Electrode-motion noise was applied at levels of 24, 12, 6, and 0 dB, with 0 dB representing equal AC signal and added-noise power. For comparing combinations of methodologies against one another, combined minus confidence-adjusted class-balanced retained error was computed at nominal 90% matched coverage and averaged equally across the four noise levels. A claim of superiority also required a favorable 95% interval excluding zero and at least 80% retention of the positive class at every severity. Results: Testing consisted of 2,085 ECGs from 1,812 participants, including 151 AF-positive recordings from 129 participants. There was a primary difference of -0.26 percentage points (95% paired bootstrap confidence interval, -2.70 to 1.14). Across the four noise levels at nominal 90% matched coverage, AF retention ranged from 9.27% to 48.34%, failing the prespecified criterion. Indeed, no true positive classifications existed within retained samples at 24 and 12 dB. Similarly, in an exploratory sample of 1,864 ECGs, no advantage of the combined approach could be determined. Excluding the predicted class in ranking by quality resulted in an average mean balanced retained error rate of 16.10% and retention of 85.43% to 89.40% of AF instances. This constituted a marked improvement upon the initial quality-plus-class ranker whose balanced retained error was 50.00%. Conclusion: Waveform quality measures failed to demonstrate superiority relative to the specified methodology in the presence of added noise. Class-level coverage revealed an important deficiency that overall coverage would have missed entirely. The result without predicted class deserves a closer examination.

Keywords: electrocardiography; atrial fibrillation; signal quality; selective prediction; uncertainty; class retention; noise robustness

Introduction

Interpretation of ECGs via automated classifiers results in a classification output and associated auxiliary data, which may be used to gauge the confidence of a particular classification result. While these issues overlap, it is critical to distinguish between the two: a good classifier can produce accurate classifications on average while still failing in individual instances. Another important consideration is the effect of contaminating signals: here, the probability returned by a classification model does not correspond to the fidelity of the waveform analyzed.

The solution proposed here involves selective classification, whereby the classifier chooses not to return a classification for particular instances. Typically, the performance metric considered in selective classification studies is the error rate on the retained classifications versus the fraction retained [1]. There exists substantial

literature on selective prediction for ECGs. Indeed, Zhang et al. investigated uncertainty-based selective prediction via Bayesian neural networks and MC dropout on 12-lead ECGs [2]. Note, however, that no claim is made regarding the novelty of selective prediction itself in this manuscript. Instead, the goal is to determine whether auxiliary waveform quality information helps order error rates for records subject to new kinds of contamination beyond that considered during ranking model development.

Another problem occurs regardless of the retained classification error: what records did the model choose to retain? For an unbalanced AF annotation task, a classifier could retain nearly all negatives but reject most positive records. Thus, the total retention rate might still be relatively high despite very few retained positives. Meanwhile, the sensitivity measured only on retained positives would be favorable despite deferring most actual positives. These metrics capture orthogonal properties of the same decision rule and should both be reported.

Here, we utilize the PTB-XL resource, its participant partitioned cross-validation folds, select a single lead recording, and create an additive-noise benchmark dataset. Error-ranker development used baseline wander and muscle artifact contaminants, while electrode motion noise was left as the test contaminant. Two competing approaches were compared: a confidence-adjusted learned error ranker, and a supplemental waveform quality-based error ranker using seven features alongside the original ranker. The core hypothesis tested is that the hybrid error ranking scheme reduces balanced error rates relative to the learned ranker alone at a fixed retention rate budget. To qualify as positive, the method must yield a minimum level of retention on actual AF cases. Other analyses were performed on another ECG source, multiple training seeds, ablation experiments, and subgroup retentions. The objective of this work is not clinical decision support nor consumer wearable validation; instead, the goal is a reliability benchmark comparison study.

Methods

Study design, data sources, and protocol history

Herein, we perform a second analysis of publicly available, deidentified electrocardiography (ECG) signals. Our primary dataset was PTB XL version 1.0.3 obtained from PhysioNet [3, 4, 9]. Our reference task entailed predicting whether the signal was associated with the AFIB rhythm category annotation compared to another explicitly annotated rhythm. Our prediction unit was the recording, whereby individuals could provide multiple recordings. There was participant-disjoint assignment in all five analytic roles.

Our finalized protocol was locally hash-locked on 15 October 2026 ahead of acquiring our final-test waveforms and predicting primary electrode motions. We maintained our dated draft, prospective clarification notes, model locks, and amendment history record. This work did not represent public preregistration. An implementation fix updated the floating-point accuracy of the clean reference used for verification of our added-noise SNRs. This change did not affect our waveform generations, model training, feature extraction, thresholds, nor statistics and came ahead of our primary predictions of electrode motion. Our reruns on unaltered signal waveforms agreed precisely with our previous runs.

Any external validation evaluations and robustness assessments were documented independently after observing our primary findings. Such analyses remain exploratory despite decisions being made beforehand. They neither fine-tuned nor substituted our primary approach. Predictive modeling reportability adherence was assessed with respect to TRIPOD+AI topics [8], without claiming full checklist certification.

Cohort construction and reference labels

For inclusion, an age of at least 18 years, a human-review true provider flag, and one or more rhythm statements according to the provider dictionary were required. Records were considered positive if they had an AFIB key in the SCP annotation dictionary, regardless of the numeric probability associated with that key. Conversely, records having rhythm statements but lacking an AFIB key were labeled negative. This criterion ensures that unknown

rhythm probabilities were not treated as non-annotation. Note that negative annotation does not indicate clinical negativity of AF based on the ECG alone.

Coannotations of other rhythms were preserved and flagged for an exclusion sensitivity analysis. Other categories not considered predictors included diagnostic statements, narrative report text, review flags, participant identifier, fold identifier, and native noise annotations. Record ID numbers served merely for traceability purposes, noise labeling, and enforcing a deterministic tie-breaking rule for ranking. Our final analysis cohort therefore consisted of 15,292 records of 13,796 participants. For training purposes, folds 1 to 6 were used. Checkpoint and classification-threshold choice used fold 7, ranking fit used fold 8, and unlabeled policy selection used fold 9. The test set comprised fold 10 (Table 1).

Reference annotations were accessed during the process of determining eligibility. Once this process was completed, the fold 9 outcome labels were neither incorporated into the policy arrays nor utilized for threshold choice. Test outcomes were used exclusively to evaluate predictions of locked classifiers. These constraints define our analytical procedure, but should not be interpreted to suggest that any source metadata remained encrypted and inaccessible.

Signal handling and baseline classifier

From among the 100 Hz, 12-lead records provided by PhysioNet, Lead II was selected based on its header designation. Each record contained 1,000 samples. Digital checksums, layout, gain, baseline correction, physical unit conversion, and finiteness checks were performed prior to processing. All selected header-signal pairs agreed with those provided by PhysioNet based on their SHA-256 hash values. Furthermore, each record requested was accounted for if it could not be included in the study. Exact Lead II waveform hashes showed no identical-waveform overlap between analysis roles, representing an additional integrity verification method aside from participant separation.

Our classifier consisted of five 1D convolutional layers with widths of 16, 32, 64, 64, and 64. For all layers, kernel sizes were 7, strides were 2, padding was 3, along with batch norm and ReLU activation functions. Afterward, global average pooling, dropout 0.2, and a final linear output layer comprised the remainder of the network architecture. Training was accomplished using binary cross-entropy with logits, using the AdamW optimizer, with a learning rate of 0.001, weight decay of 0.0001, a batch size of 64, and a maximum of 60 epochs. Early stopping involved a patience value of 8 and minimum loss improvement of 0.0001.

First, each record was mean-centered and standardized to one standard deviation as estimated solely from training records. Standardizing to unity variance per record was avoided to retain inter-record differences in amplitude scaling. The best performing checkpoint in terms of fold 7 binary cross-entropy loss was retained. Classification of AFIB was set according to the highest confidence threshold producing at least 90 percent AF sensitivity across all records in fold 7 without modification. The fixed primary seed was 20261015. Our classifier was fitted on folds 1 to 6 and was not refitted on folds 7 to 9. The feasibility of developing our model involved achieving an AUROC of at least 0.80 and a minimum of 0.50 specificity on our checkpoint set (at the chosen operating point). These criteria did not imply any validation claims whatsoever. We made use of a fixed RR feature logistic regression baseline and negative reference comparator solely during development.

Recorded noise and withheld-source evaluation

Noise stimuli came from the MIT-BIH Noise Stress Test Database, version 1.0.0 [5]. Baseline wander (bw) and muscle-artifact (ma) noises were used for training rankers and selecting policies, but electrode motion (em) waveforms were left out of both tasks, evaluating instead transfer to the withheld em source, which also contains other artifact types (as noted in Ref. [5]).

Each source yielded two channels. For each ECG and family type, SHA-256 hashing deterministically chose a channel and a 10-second window within each source for analysis purposes. Training data consisted of seconds 0

to 580, policy selection data came from seconds 600 to 1180, and test data covered seconds 1200 to 1780. No windows overlapped in roles. Multiple ECGs could have shared the same window, and that particular window was always used regardless of severity. Each family member used one noise stimulus, rather than several donors.

Each 10-second noise window was independently resampled at 100Hz via polyphase filter factors of 5/18, applying a Kaiser window with beta equal to 5 and padding with zeros. Noise was added to the centralized ECG after scaling the AC RMS to achieve a target signal-to-noise ratio (SNR), which could take values of 24dB, 12dB, 6dB, or 0dB. The reported SNR here refers specifically to the AC signal-to-added-noise power ratio within the final analyzed 100Hz signal. It does not refer to signal quality overall, nor does it follow the WFDB QRS amplitudes. Artifact could also exist in original ECG signals. Powerline interference and dropout simulation were not performed.

Waveform-quality measurements and error rankers

To summarize the results here, the waveforms' measurements were AC STDDEV, RANGE, NFA, rmsADSTDEV/ACSTDDEV, LPF-PWR, HPF-PWR, and Pearson kurtosis. They are numerically defined in Supplementary Table S2. All of these inputs went into an error-ranking model without accompanying any clinically validated acceptability threshold for any particular measurement.

In the error-ranking model, there were three logistic classifiers, all using L2-penalized logistic regression with C=1. The first compared confidence-adjusted predicted AFIB probabilities—the predicted probability of the event AFIB and the absolute distance away from the fixed AFIB classification cutoff, and the predicted AFIB class. The second included the seven waveform measurements, their associated missingness flags, and predicted AFIB class. The third, the combination model, incorporated the full set of 17 inputs: those from the previous two, including the predicted class itself. Therefore, in the terminology of the study, the quality-only indicator means excluding probability-related variables and not the predicted class itself. Unadjusted confidence was calculated as one minus the greater class probability.

Error-ranker fitting occurred only with fold 8; final testing used fold 10. The 1,405 raw recordings along with their respective bw/ma transformations totaled nine variants per recording—eight transformed versions, plus one original. Each original recording had a combined fitting weight of 1 across its nine variants. For imputing missing median statistics and normalizing feature scaling parameters, fold 8 data alone was used, where a feature whose value was entirely null was imputed as 0. Logistic regression parameters were set as follows: C=1, fit_intercept, solver=lbfgs, max_iter=2,000, tol=0.0001, and no class weights were applied. There were 393 errors among the 9 x 1,405 total waveforms. The minimum feasible participant count for inclusion criteria is 30 unique patients having any incorrect variant and 30 unique patients with any correct variant; these values observed were 79 and 1,271, respectively. These numbers do not represent disjoint sample sizes.

Coverage rules, primary endpoint, and AF retention

ECGs were retained for matching with regard to the primary budget-comparison metric based on the lowest scores within each severity category, retaining ceil(0.90 x N) ECGs. For ties, resolution was based upon the lexicographical order of the ECG identifiers. Class labels played no role in selection. Balanced error was one half of the retained false-negative rate plus one half of the retained false-positive rate. Denominators were retained counts of each respective class. Effect magnitude was the average difference in balanced error between combinations versus adjusted confidence. The four EM severities had equal weighting. Values less than zero were indicative of advantages to combination. Error values were undefined when one or both retained classes were empty; missing data were never assumed to be zeros.

Retention rates of cases positive for atrial fibrillation (AF-case retention rate) referred to proportions retained among positive cases. Sensitivity was calculated as true positives divided by retained positive cases. The preservation rate of true-positive classifications divided by original positive cases. For clarification, see supplementary Table S3.

Superiority was claimed if both criteria were met: a favorable 95% confidence interval below zero; ≥80% case retention in each primary severity level. These thresholds of 80%/90% were researcher-designated rather than clinically validated; furthermore, adequate power was not established. Corrupted ECG copies contributed no additional participants.

Fixed score thresholds for deciding retention or deferral were chosen among those scores without any label information on fold 9. For each type (unmodified/bw/ma) of ECG and each noise level, weights were 1/3 and 1/12, respectively. Thresholds were set according to the minimum score having a cumulative probability of at least 0.90, implying a retention rate of scores ≤ threshold. A tied threshold could raise coverage beyond target levels. These parameters were fixed, and test outcome-dependent adjustments were prohibited.

Statistical inference and exploratory analyses

The primary bootstrap confidence interval estimation procedure is performed on 2,000 pairs of participant bootstraps with seed 20261018 using percentile boundaries of 2.5% and 97.5%. Recordings and severity variants associated with a participant sample remain intact throughout. Methods employ identical sampling with replacement procedures, ranking the samples' retained recording budgets separately for each bootstrap draw. Bootstrap duplicate-handling validation has been conducted using explicitly expanded resamples. Confidence intervals depend on the fitted model and the specified noise library; uncertainty from training set sampling and noise donors is ignored. Undefined replicates are counted and excluded from interval calculation; a numeric interval is allowed only if no more than 1% of replicates are undefined.

Severity transfer exploration used 18dB, 9dB, 3dB, and -6dB on the test participant pool for the bw, ma, and em algorithms. None of these severities was present during ranker fitting; 18dB, 9dB, and 3dB fell within the fitted severity range, while -6dB was below its lower bound. The external source consisted of the independent ECG arrhythmia 1.0.0 resource [6,7]. Prior to label and age reading, 2,000 record path samples are chosen by fixed hashing order out of its total 45,152 records. Requirements of adult age, one out of eleven rhythm code types, and signal compatibility were imposed, without replacing excluded records. AFIB and AF (atrial flutter) diagnoses were discriminated separately. Resampling of external records to 100Hz lead II utilized native headers, a factor of 1/5 and Kaiser window beta parameter of 5 with constant padding interpolation. Model thresholds and normalizations are retained. Resampling treats each patient file separately in accordance with the provider's one-record-per-patient policy; linkage information does not exist.

Robustness experiments repeat the developmental process using seeds 20261016 and 20261017. Seven predefined ablations separately remove input groups or predicted-class terms, along with evaluations at review budgets of 80%, 90%, and 95%. Mixed rhythms are also removed from analysis. Only one ECG is kept from each participant, selected as the earliest recording available. Eligible patient age, recorded sex, and native artifact presence are considered stratification parameters. At least 30 positive and 30 negative distinct participants must occur within a stratum for estimates. Ranking and matched coverage are recalculated within each stratum; coverage uses all original stratum records as its denominator. Native artifacts are those whose baseline drift, static noise, burst noise, or electrode problems attributes contain content, while extra beats and artificial pacemaker signals are not noise. Exploratory intervals are pointwise descriptive only without multiplicity-adjusted hypothesis testing.

Results

Cohort and unmodified-signal classifier performance

All of the chosen primary-source files passed integrity checks. Our test consisted of 2,085 ECGs drawn from 1,812 individuals, including 151 AF positives and 1,934 negatives. The primary prevalence rate of atrial fibrillation based on these records was 7.24%. The primary CNN checkpoint occurred after epoch 8, with a

classification threshold set to 0.562615. A checkpoint AUROC value of 0.9898 was achieved using fold 7 records; nevertheless, they had already been used to determine both selection and thresholds.

Based solely on independent test ECGs with no modification whatsoever, the AUROC was 0.9772, average precision 0.8352, Brier score 0.0213, sensitivity 80.13%, and specificity 98.66% (Table 2). Meeting our 90% sensitivity criterion on checkpoint data did not ensure meeting it on test data. When applying recorded additive electrode-motion noise, we noted that balanced error rate increased from 11.22% (24dB) to 25.18% (0dB). Meanwhile, AUROC dropped from 0.9763 to 0.9185. Among incorrect predictions under various levels of em noise, the average maximum-class probability value ranged from 0.72 to 0.79. This represents descriptive confidence, not proof of properly calibrated probabilities for error.

Primary comparison and class-selection failure

Using any of these approaches at the target 90% nominal coverage level retains 1,877 out of 2,085 ECGs, corresponding to an average actual coverage level of 90.024%. The difference between the combined and confidence-adjusted mean balanced retained error equals −0.2635 percentage points (95% confidence interval [CI] −2.7034, +1.1410). No cases lacked primary bootstrap replicate values. The confidence interval encompasses improvements and worsened performance alike and did not satisfy the predefined superior criterion (see Table 3 and Figure 1).

Retention failure with AF positives was considerable. Combined retention rate of AF positives equaled 14, 17, 32, and 73 at noise levels 24, 12, 6, and 0dB, respectively. AF retention rates equalled 9.27%, 11.26%, 21.19%, and 48.34% at these severities, all falling well below the 80% threshold. Furthermore, retention rates at 24 and 12dB were so poor that all retained positive cases became negative classifications. As such, the number of retained true positives equaled zero. The balanced retained error was 50.00% at 24 and 12dB, compared to the unfiltered estimate of 11.22% and 12.57%, respectively. Contrastingly, ordinary retained errors were just 0.75% and 0.91%, respectively, owing to the preponderance of retained negative cases. This shows clearly how these measures differ. Using raw confidence resulted in lower average mean balanced retained error (13.00%) across primary severities, but retained 37.75% to 56.29% of the AF positives. Raw confidence did not meet the case-retention threshold either. None of these facts changes the predefined comparison. Coverage curve estimates and fixed-validation estimates are available in supplementary material.

Severity transfer and sampled external evaluation

The new measures of internal EM severity resulted in an average difference between the combined ranker minus adjusted ranker error rates of -0.27 percentage points (95% CI: -2.41, +1.45). While the combined ranker satisfied the criterion of AF retention at -6 dB at an AF retention rate of 85.43%, the combined ranker failed the criterion for 18 dB, 9 dB, and 3 dB. This result did not contradict the previous result.

The external validation set contained 1,864 eligible ECGs: 103 AF+ cases and 1,761 AF- cases. From the 2,000 chosen header files, 136 were removed for failure to meet either age (N=98) or absence of eligible ECG rhythm coding (N=40) requirements; two header files met both exclusion criteria. All selected header files matched provider-provided checksum values and eligible signals matched selected header files. Exact external waveforms duplicates or overlapping waveforms with respect to original samples could not be found, yet these results do not indicate patient distinctiveness alone.

On the external ECGs without modification, the unaltered classifier obtained an AUROC of 0.8879, average precision of 0.2114, Brier score of 0.1397, sensitivity of 81.55%, and specificity of 82.34%. For the difference in error rate between combined minus adjusted classifiers, the average external difference was +0.39 percentage points (95% CI: -0.56, +2.01) with AF retention of 71.84%. Over various external EM severities, the average difference was +0.35 percentage points (95% CI: -0.45, +1.65), and the AF retention varied from 66.99% to 69.90% (Table 4; Figure 2). None of the external summaries dominated.

Training-seed sensitivity and feature ablations

Ablation of the predicted AFIB class resulted in a new configuration having 14 quality/missingness features and no predicted AFIB status (hereafter "quality"). For quality, the average balanced retained classification error was 16.10% compared to 50.00% for the original quality-plus-class configuration. The difference in mean errors was thus –33.90 percentage points with a 95% pointwise confidence interval of [-37.26,-30.42]. The values for AF retention were 85.43%, 86.75%, 87.42%, and 89.40%, which exceeded the threshold at each respective original event severity level (Table 5; Figure 3).

Comparison against which ablation was conducted is critical here. While 33.90 percentage points improvement sounds impressive, it was against the quality-plus-class ranker. When classification was done without filtering (henceforth, "unfiltered"), the mean balanced error was 16.66% while retaining 100% of all instances. Raw prediction confidence had a mean error rate of 13.00%, though at significantly lower retention rates (Figure 3). Clearly, the dramatic contrast achieved via feature removal did not lead to an equivalent benefit when comparing against unfiltered classification, nor did it yield an overall superior performance than any possible configuration thereof. Indeed, the difference from the full combined ranker after removing predicted class was –0.42 percentage points with a 95% confidence interval of [-2.54,+2.39] (with AF retention rates from 9.27% to 43.71%). All six corresponding 95% intervals included zero. Furthermore, we note explicitly that quality ablation used the original classifier seed, not tested against other seeds or even a separate external validation set.

Cohort, subgroup, and review-budget checks

Without considering the mixed-rhythm category, there were 2,071 electrocardiogram (ECG) recordings left, which gave a combined minus adjusted difference of –0.61 percentage points (95% CI: –2.63 to +0.95). Restricting the analysis to the earliest recording by each subject resulted in 1,812 ECG recordings, with a difference of +0.23 percentage points (95% CI: –0.88 to +0.99). Neither of these analyses included re-estimation of the original models.

The age stratum 18 to 64 included 1,097 recordings but had only 25 AF-positive participants. Hence, this group had fewer than the required minimum number of 30; hence, performance was deliberately avoided. Age ≥65, including the oldest-aged provider-deidentified code, native artifact strata, and recorded-sex groups 0 and 1 met the requisite threshold. For all combinations of the estimated subgroups, the confidence intervals overlapped zero, and none passed the every severity retention criterion (refer to Supplementary Table S6).

Annotations of native artifacts were not necessarily confined to Lead II alone. Further, absence of an annotation did not establish absence of artifact.

The budget-matching and validation-fixing strategies were each assessed separately. Positive cases consisted of deferral of AF cases, deferral of true-positive classification, retention of false-negative cases, and errors, where there is a maximal classification probability ≥0.90. Both strategies distinguished sensitivity among retained positives and preservation of original positive cases and identified no replacement primary strategy.

Reproducibility and computational measurements

Automated audits of the final codebase revealed 64 completed unit tests. An examination of 34,454 waveforms/noise files found them correctly verified by saved provider checksums, confirming that saved classifier probability predictions and rankings had been successfully reproduced. Metrics of retained set performance and implementation correctness were independently verified through calculation on reference samples, as well as the correctness of the bootstrap procedure itself. All original raw data, fitted models, and prior results were stored verbatim, bit-for-bit, making up internal computational checks but no human peer review.

Running on an AMD EPYC CPU, using 2 Torch threads, the per-single-record medians for CNN inference were 0.786ms, seven quality measurements were 1.055ms, and loading/scoring of a JSON file was 0.175ms. Corresponding 95th percentiles were 1.709ms, 1.272ms, and 0.228ms, respectively. This timing analysis sampled

the first 100 test ECGs without modification after a warm-up batch. Measurement does not account for acquiring a 10 second ECG recording. Measurement does not verify latency and safety requirements at a device level. Per-process peak memory during loading and evaluation was 503.6MiB, accounting for all total memory consumption, including non-model overheads.

Discussion

The inclusion of waveform quality did not establish improved performance on the withheld source of noise. Combined approach met nominal 90% overall retention but did not satisfy the 80% AF-case retention requirement at any primary severity level. Rather than concluding with a small difference in average retained error, the critical observation was that a system retaining ≥90% could retain virtually no true-positives of AF diagnosis at two severity levels. This highlights how the reliance on overall coverage as a measure of reliability can lead one astray.

Indeed, this case serves to demonstrate the importance of considering retained class balance and errors. For instance, 24 dB produced a retained set containing only 14 AF-positive recordings, each classified erroneously as negative. Average retained ordinary error was ≤1%, while balanced retained error was 50%. The former was calculated on a set consisting primarily of negative cases. Ignoring the latter statistics and solely focusing on averages would result in failing to recognize the absence of positive cases.

It is tempting to interpret the above result as implying that deferral led to missed clinical diagnosis. However, no downstream review process or alternative recordings were simulated to assess such impacts. While deferred recordings may receive proper focus in an optimized workflow design, the current study could not presume such outcomes. Instead, the experiments show how outputs would be modified according to retention rules. Any follow-up experiments must consider the consequences for deferred cases and workloads therein.

Ablation analysis shows that the contents of a quality ranker can carry weight alongside computational complexity. The version of the quality ranker without predicted class retained considerably more positive cases and showed significantly lower balanced retained error compared to the original ranker, despite similar computations. This provides an interesting sensitivity comparison since the waveform measurement and fitting components were unchanged. While development-specific effects on predictive accuracy could influence rankings, evidence alone cannot support that claim here. Furthermore, removing prediction class from the combined model failed to rectify failures in that method, meaning that class dependence was insufficient for observed performance degradation in general.

The value of this finding lies in limitations of comparison that reveal itself explicitly. 16.10% class-balanced error with high retention rate was quite similar to 16.66% achieved by the unfiltered classification approach, while the confidence-only ranker improved error rates at cost of greater deferrals of positive cases. These results highlight separate strategies trading off between retained errors and cases. They do not justify selection of the better performing ablation as an apparent 33.90-point improvement, nor labeling of confirmation experiments thereafter. An additional round of evaluations is needed for a new model.

Both the external sample evaluation and alternative classifier-training seeds could not reverse the conclusion made from combined approaches. The unchanged classifier performed with worse discrimination on external records than the unmodified PTB-XL test set, but differences arise from multiple considerations including populations, rhythm coding standards, noise prevalence, and preprocessing procedures. The test records came from an external source, while the augmented noise libraries were retained. This describes a bounded transfer assessment without establishing a combined-method advantage.

The presented benchmark study complements prior efforts in ECG uncertainty rejection (2). It considers composition of retained records under a given coverage budget with a withheld source of noise and utilizes established confidence measurements with waveform features rather than proposing a novel framework for

measuring uncertainties. The principal methodology lesson is clear. When assessing a deferral strategy, one must ensure that the process retains cases that a classifier aims to classify.

Limitations

In contrast to previous works, our target objective was retrospective AFIB annotation in a short resting ECG rather than adjudicated longitudinal diagnoses or prospective screening outcomes. Other annotated rhythms made up the negative set; comorbidities and labeling errors can complicate any interpretations.

We extracted a single lead from recordings made with 12 leads. They were not collected using wearable devices; we retained the processing performed by the original provider, and synthetic noise cannot emulate each possible type of electrode dropout, movement, malfunction, and so forth. Our withheld EM source contains diverse types of artifacts. While the SNR pertains to added noise, the real challenge lies elsewhere.

Windows within each noise type came from individual recordings, with windows reused among recordings of ECGs. Test assignment used 116 combinations of channel–window pairs within each noise family type, where up to 36 ECGs used common windows for EM. Non-overlapping role windows minimized direct resampling of data segments between development and testing phases but did not provide completely disjoint donor pools. Participant bootstrapping intervals conditioned on those libraries and learned classifiers.

Our experiments employed one CNN architecture and one primary seed, along with fixed error-ranked simple logistic classifiers. Sensitivity seeds alone fail to replicate the whole classifier family properly; no formal evaluation was undertaken regarding adequacy of sample sizes. Spectra for 10 second signals suffer poor frequency resolution in estimating low-frequency powers.

The external analysis considered 1,864 eligible records within the prespecified subset of 2,000 headers in our external database; we did not analyze the entire database content. Patient intervals depend on the one-record-per-patient assertion by the provider; independent verification was impossible due to lack of identifiers. Direct inspection of waveforms is insufficient for confirming patient identity.

We performed further exploration analyses based on primary findings. The promising quality ablation was not tested in new data or with extra classifier seeds; moreover, confidence intervals were determined pointwise without multiplicity correction. Performance in the age 18-64 stratum was not estimated owing to insufficient AF-positive participants; descriptive coverage checks are inadequate for assessing demographic balance.

It should be mentioned explicitly that the retention threshold represents a researcher choice; we evaluated none of the clinical workflow steps, user interfaces, hardware implementations, or benefits to users' well-being. The local experimental protocol lock and automated audit procedures conducted internally cannot serve as pre-registration or external peer review. While the frozen technical record ensures computational reproducibility, author validation and scientific evaluation must follow independently.

Conclusions

The superior performance expected from the combined error ranker compared to the confidence-adjusted baseline according to the waveform quality was not realized in the present benchmark of the single lead ECG. Although high overall retention rates hid extremely poor retention rates for the AF-positive case, resulting in a total loss of retention of correctly classified cases at two levels of noisy ECG signals, a simple quality-based ranking system, independent of predicted classes, gave rise to a more favorable case-retention tradeoff than the original quality-plus-class ranker. Nevertheless, the exploratory ablation requires a fresh fixed evaluation. Future studies must disclose both the error and retained cases.

Declarations

Ethics and scope

All analyses used publicly available retrospective data that has been anonymized as permitted under the terms of their respective source licenses, with no recruitment of participants nor interaction with patients in a clinical setting. No new institutional approvals or exemptions are claimed here. Source study approvals are specified as provided by the data providers. Submission-specific requirements must be confirmed before submission.

Data, code, and protocol availability

The source material and the respective URLs to obtain particular versions are provided in the references. PTB-XL and the additional data source both license their material under CC BY 4.0, whereas the NSTDB uses the Open Data Commons Attribution License v1.0. The full technical archive contains source material listings, role lists for participants, model fits, parameterizations, prediction tables, summary statistics, testing, and evidence freezing. An abridged version, the analysis archive, lacks the source data from cohorts and external sources and comes equipped with guidelines for regeneration. At present, this archive remains private, and no public ECG research repository or clinical offering has been released for this project.

AI assistance

GPT assisted with code development and debugging, analysis execution and checks, and manuscript drafting and editing. The author must verify the methods, results, references, and final wording before submission.

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Table 1. Participant-disjoint analytic roles

Role	Folds	ECGs	Participants	AF+ ECGs
Classifier training	1,2,3,4,5,6	8,301	7,555	510
Checkpoint / cutoff	7	1,410	1,275	90
Error-ranker fitting	8	1,405	1,291	91
Unlabeled policy selection	9	2,091	1,863	Not used
Final test	10	2,085	1,812	151

Policy-selection labels were not used or exported. Test AF count is 151 after the locked evaluation. Total: 15,292 ECGs and 13,796 participants.

Table 2. Classifier performance before deferral

Condition	AUROC	AP	Brier	Sens. %	Spec. %	Bal. error %
Unmodified	0.9772	0.8352	0.0213	80.13	98.66	10.61
em 24	0.9763	0.8323	0.0214	78.81	98.76	11.22
em 12	0.9711	0.8041	0.0238	76.16	98.71	12.57
em 6	0.9596	0.7343	0.0307	66.23	98.40	17.69
em 0	0.9185	0.5879	0.0464	51.66	97.98	25.18

All 2,085 test ECGs contribute in each row. AP is average precision. Added-noise SNR is stated in dB; unmodified does not imply artifact free.

Table 3. Primary matched-budget results under withheld em noise

SNR dB	Adjusted error %	Combined error %	Diff. pp	AF retained	AF retained %	Retained TP
24	50.00	50.00	+0.00	14/151	9.27	0
12	50.00	50.00	+0.00	17/151	11.26	0
6	46.55	43.80	-2.75	32/151	21.19	4
0	24.01	25.70	+1.69	73/151	48.34	36

Both methods retain 1,877 ECGs at every severity. Primary mean difference: -0.2635 pp; 95% CI -2.7034 to +1.1410. Negative favors combined. Every combined AF-retention value fails the 80% research criterion.

Table 4. Exploratory transfer summaries

Evaluation	Difference pp	95% interval pp	Combined AF retention %
Internal em 18/9/3/-6 dB	-0.27	-2.41 to +1.45	9.93-85.43
External em 24/12/6/0 dB	+0.35	-0.45 to +1.65	66.99-69.90
External unmodified	+0.39	-0.56 to +2.01	71.84-71.84

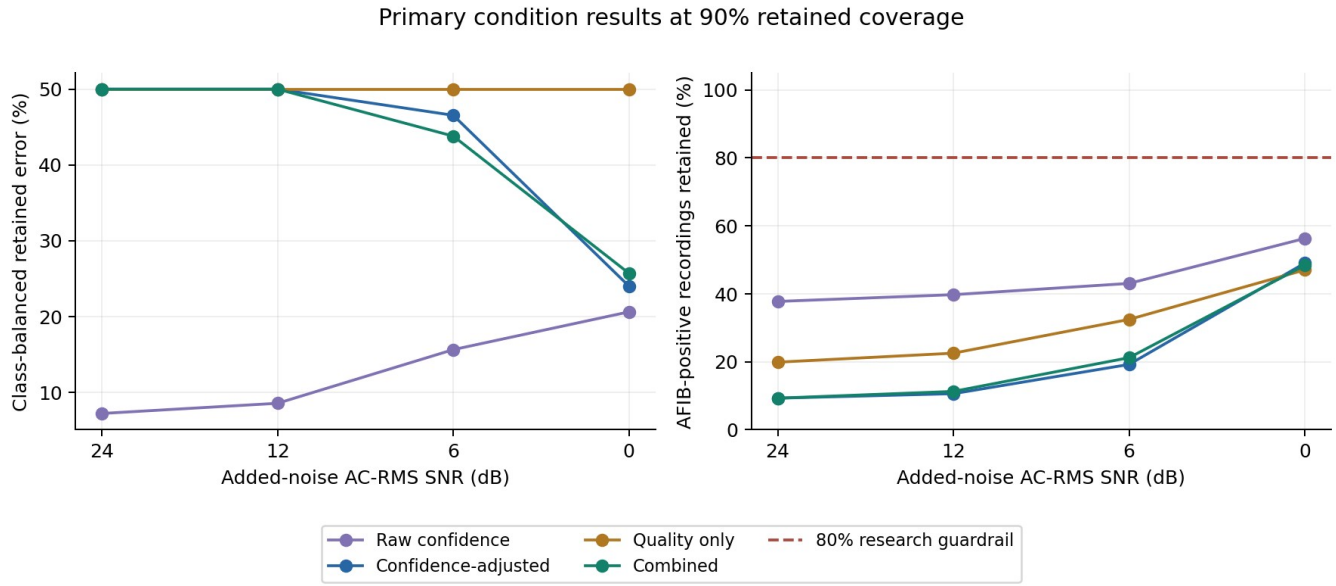
Same primary models and policies; no refitting. External results use a fixed eligible sample of 1,864 records. Intervals are pointwise; external resampling assumes one record per patient.

Table 5. Error and case-retention tradeoffs

Method	Mean balanced error %	AF retention range %
Raw confidence	13.00	37.75-56.29
Confidence-adjusted	42.64	9.27-49.01
Quality + predicted class	50.00	19.87-47.02
Combined	42.38	9.27-48.34
Quality without predicted class	16.10	85.43-89.40
Unfiltered classifier (100% coverage)	16.66	100.00

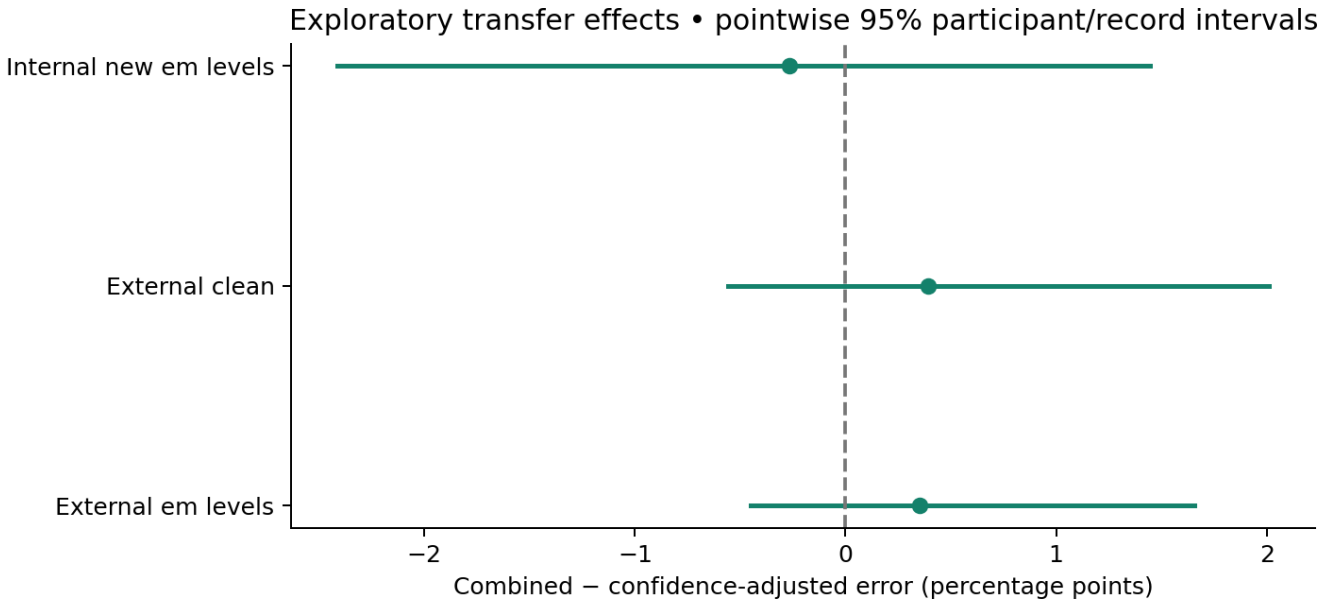
Rankers use nominal 90% matched coverage. Unfiltered classification uses 100% coverage. Quality without predicted class is a post primary exploratory ablation. Its -33.90 pp contrast is against quality + predicted class; no interval for its difference versus unfiltered classification was established here.

Figure 1. Primary error and AF retention



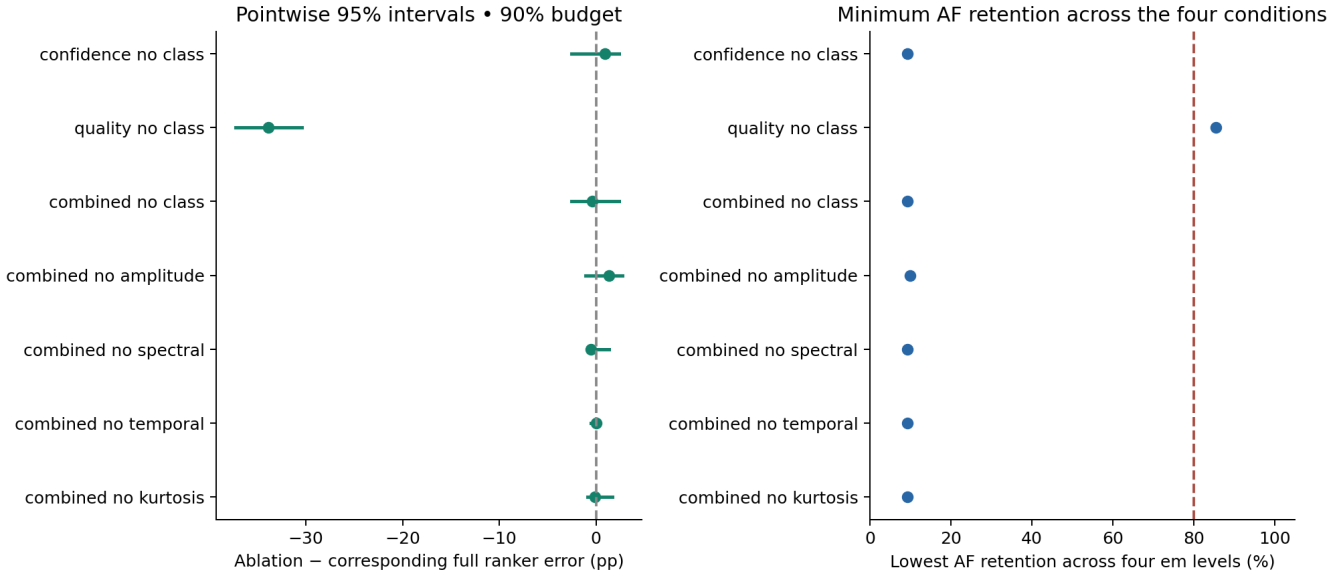
Retained balanced errors as well as atrial fibrillation (AF) cases at identical coverage rates of 90%. Each level of severity employs the same pool of 2,085 electrocardiograms (ECGs). An AF retention rate of 80% defines a scientific benchmark. The original composite technique cannot perform adequately on any level of severity; see Table 3 for specific statistics.

Figure 2. Transfer-effect intervals



The plotted difference is combined minus confidence-adjusted balanced retained error. Values below zero favor the combined error ranker. Percentile intervals are determined based on 2,000 paired participant resamples internally and paired record resamples externally. External percentiles are calculated through resampling of records, assuming one ECG per patient.

Figure 3. Exploratory feature ablations



For each ablation, performance is assessed against the corresponding original ranker—not against a common unfiltered reference model. Comparisons are made regarding the quality of predictions without the class label, versus the quality when utilizing the predicted class along with the original ranking accuracy. Point-wise confidence intervals depend upon both the classifier and noise assignment.